Practice Midterm

1. A randomly selected sample of $n = 12$ students at a university is asked, "How much did you spend for textbooks this semester?" The responses, in dollars, are

200, 175, 450, 300, 350, 250, 150, 200, 320, 370, 404, 250

(a). Construct an approximate 95% confidence interval for the population mean using the CLT.

(b). If the data are assumed from $N(\mu, \sigma^2)$, construct an exact 95% confidence interval for the population mean.

$\left[\sum_{i=1}^{12} x_i/12 = 284.9,\ s^2 = 96.1,\ z_{0.025} = 1.96,\ z_{0.05} = 1.645,\ t_{0.025,11} = 2.20,\ t_{0.025,12} = 2.18,\ t_{0.05,11} = 1.80,\ t_{0.05,12} = 1.78\right.$, where z_α is upper α quantile of standard normal distribution and $t_{\alpha,k}$ is upper α quantile of t distribution with k degrees of freedom.$\Big]$

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \dot{\sim} N(0,1)$$

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}}$$ approx. $(1-\alpha)100\%$ CI for μ.

95% CI. $\alpha = 0.05$ $z_{0.025} = 1.96$

$n = 12$ $\bar{X} = 284.9$ $S = \sqrt{96.1}$

$X_1, \cdots X_{12} \overset{iid}{\sim} N(\mu, \sigma^2)$

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \sim t_{n-1}$$

no approx.

$$\bar{X} \pm t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}}$$

exact CI

$t_{0.025, 11} \approx 2.20$

If σ is known

$$\bar{X} \pm z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim N(0, 1)$$

exact CI

2. Let $X_1, \ldots, X_8$ be a random sample of size $n = 8$ from a Poisson distribution with mean μ. Reject the null hypothesis $H_0 : \mu = 0.5$ and accept $H_1 : \mu > 0.5$ if the observed sum $\sum_{i=1}^{8} x_i \geq 8$.

(a). Compute the significance level α of the test.

(b). Find the power function $\gamma(\mu)$ of the test as a sum of Poisson probabilities.

$$\alpha = P_{\mu=0.5}\left(\sum_{i=1}^{8} X_i \geq 8\right) = P(\text{Poiss}(4) \geq 8)$$

$$\gamma(\mu) = P_\mu(\text{reject } H_0) = P_\mu\left(\sum_{i=1}^{8} X_i \geq 8\right)$$

$$= P(\text{Poiss}(8\mu) \geq 8)$$

3. Let $X_1, X_2, \ldots, X_n$ be iid $N(0,\theta)$, $0 < \theta < \infty$. Let $T_n = n^{-1}\sum_{i=1}^{n} X_i^2$.

(a) Is T_n an unbiased estimator of θ?

(b) Is T_n an consistent estimator of θ?

(c) What is the limiting distribution of $\sqrt{n}(T_n - \theta)$?

$E(T_n) \overset{?}{=} \theta$

$\underbrace{Var(X_i)}_{\theta} + (E(X_i))^2$

$E(T_n) = \frac{1}{n}\sum_{i=1}^{n} E(X_i^2) = \frac{1}{n}\sum_{i=1}^{n}\theta = \theta$

Unbiased!

$T_n \xrightarrow{P} \theta$?

$Y_i = X_i^2$. $\qquad T_n = \bar{Y}$

By the WLLN. $\bar{Y} \xrightarrow{P} EY_1 = EX_1^2 = \theta$

$T_n \xrightarrow{P} \theta$ $\qquad$ Consistent!

$$\sqrt{n}(T_n - \theta) \xrightarrow{D} ???$$

CLT: $$\sqrt{n}(\bar{X} - \mu) \xrightarrow{D} N(0, \sigma^2)$$

$$T_n = \bar{Y} \qquad EY_1 = \theta$$

$$\sqrt{n}(T_n - \theta) \xrightarrow{D} N(0, \ Var(Y_1))$$

$$X_1 \sim N(0, \theta)$$

$$\frac{X_1}{\sqrt{\theta}} \sim N(0,1)$$

$$\frac{X_1^2}{\theta} \sim \chi_1^2$$

$$X_1^2 \sim \theta \chi_1^2$$

$$Var(Y_1) = Var(X_1^2)$$

$$= Var(\theta \chi_1^2)$$

$$= \theta^2 \, Var(\chi_1^2)$$

$$= 2\theta^2$$

$$\sqrt{n}\,(T_n - \theta) \xrightarrow{D} \underbrace{N(0,\ 2\theta^2)}$$

4. Let $Y_1 < Y_2 < \cdots < Y_n$ denote the order statistic of a random sample of size n from a distribution that has pdf $f(x) = 3x^2/\theta^3$, $0 < x < \theta$, zero elsewhere.

(a). Show that $P(c < \frac{Y_n}{\theta} < 1) = 1 - c^{3n}$, where $0 < c < 1$.

(b). If n is 4 and if the observed value of Y_4 is 2.3, what is a 95% confidence interval for θ?

$$P\left(c < \frac{Y_n}{\theta} < 1\right) = P(c\theta < Y_n < \theta)$$

$$= F_{Y_n}(\theta) - F_{Y_n}(c\theta)$$

$$P(Y_n \le y) = P(X_1 \le y, \cdots X_n \le y) = \left(P(X_1 \le y)\right)^n$$

$$= \left[\int_0^y \frac{3x^2}{\theta^3}\,dx\right]^n = \left(\frac{y^3}{\theta^3}\right)^n \qquad 0 < y < \theta$$

$$F_{Y_n}(\theta) - F_{Y_n}(c\theta) = 1 - c^{3n}$$

$$P\left(c < \frac{Y_n}{\theta} < 1\right) = 1 - c^{3n}$$

$$P\left(Y_n < \theta < \frac{Y_n}{c}\right) = 1 - c^{3n}$$

$n = 4.$ $\quad 1 - c^{3n} = 0.95 \quad c = (0.05)^{\frac{1}{12}}$

$$\left(Y_n, \frac{Y_n}{c}\right): \quad \left(2.3, \frac{2.3}{(0.05)^{\frac{1}{12}}}\right)$$

95% CI for θ!

5. A single observation of a random variable having a uniform density on $[\alpha, \beta]$ with $\alpha = 0$ is used to test the null hypothesis $\beta = \beta_0$ against the alternative $\beta = \beta_0 + 2$. If the null hypothesis is rejected if and only if the random variable takes on a value greater than $\beta_0 + 1$, find the probabilities of type I and type II errors.

$X \sim unif[0, \beta]$

$H_0: \beta = \beta_0 \qquad H_1: \beta = \beta_0 + 2$

$C = \{x : x > \beta_0 + 1\}$

$X \sim unif[0, \beta_0]$

$$\alpha = P_{\beta=\beta_0}(X > \beta_0 + 1) = 0$$

$X \sim unif[0, \beta_0 + 2]$

$$\beta = P_{\beta=\beta_0+2}(X < \beta_0 + 1) = \int_0^{\beta_0+1} \frac{1}{\beta_0+2}\,dx$$

$$= \frac{\beta_0+1}{\beta_0+2}$$

6. Let $X_1, \ldots, X_n$ be a random sample from pdf $f(x) = 5x^4$, $0 < x < 1$, zero elsewhere. Let $Y_1 = \min\{X_1, \ldots, X_n\}$. Find p so that $Z_n = n^p Y_1$ converges in distribution.

cdf of Z_n:

$$P(Z_n \le z) = P(n^p Y_1 \le z) = P\left(Y_1 \le \frac{z}{n^p}\right)$$

$$= 1 - P\left(Y_1 > \frac{z}{n^p}\right)$$

$$= 1 - P\left(X_1 > \frac{z}{n^p}, \ldots, X_n > \frac{z}{n^p}\right)$$

$$= 1 - \left[P\left(X_1 > \frac{z}{n^p}\right)\right]^n$$

$$= 1 - \left(\int_{\frac{z}{n^p}}^{1} 5x^4 \, dx \right)^n$$

$$= 1 - \left(1 - \left(\frac{z}{n^p} \right)^5 \right)^n$$

$$= 1 - \left(1 - \frac{z^5}{n^{5p}} \right)^n$$

$5p = 1.$ $p = \frac{1}{5}$

$$= 1 - \left(1 - \frac{z^5}{n} \right)^n \longrightarrow \boxed{1 - e^{-z^5}}$$

$\underline{\text{cdf}}$

$z > 0$

7. Let $X_1, \ldots, X_n$ be a random sample from Bernoulli(p). Find the approximate distribution of $Y_n = \sin(\pi \bar{X}_n)$ from the Delta-method.

$X_1, \ldots, X_n$ iid Bernoulli(p)

By CLT: $\sqrt{n}(\bar{X} - p) \xrightarrow{D} N(0, p(1-p))$

$g(x) = \sin(\pi x)$

$g'(x) = \pi \cos(\pi x)$

By Δ-method.

$\sqrt{n}(g(\bar{X}) - g(p)) \xrightarrow{D} N(0, g'(p)^2 p(1-p))$

$$\sqrt{n}\left(\sin(\pi \bar{X}_n) - \sin(\pi p)\right) \xrightarrow{D} N\left(0, \frac{\pi^2 \cos^2(\pi p)}{p(1-p)}\right)$$

when n is large,

$$\sqrt{n}\left(\sin(\pi \bar{X}_n) - \sin(\pi p)\right) \overset{\cdot}{\sim} N\left(0, \frac{\pi^2 \cos^2(\pi p)}{p(1-p)}\right)$$

$$Y_n = \sin(\pi \bar{X}_n) \overset{\cdot}{\sim} N\left(\sin(\pi p), \frac{\pi^2 \cos^2(\pi p)\, p(1-p)}{n}\right)$$